Supplementary Material

Width Confounding in Label-Free Cross-Architecture Model Selection: Diagnosis and Subspace Correction

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Appendices A–G accompany the main paper. Reference numbers refer to the main paper's bibliography.

A Marchenko–Pastur and Isotropy

Proof of Lemma 1. Write $r_{\text{eff}}(H) = \exp\big(H_{\text{ent}}(\mathbf{p})\big)$, where RankMe uses $p_i = \sigma_i / \sum_j \sigma_j$. Let $\lambda_i = \sigma_i^2/(n\sigma^2)$ and let the empirical measure include all d eigenvalues, including the zero eigenvalues when $d > n$. By the Marchenko–Pastur law [36], this measure converges almost surely to μ_γ. Define $S_d = d^{-1}\sum_j \sqrt{\lambda_j}$ and $m_{1/2} = \int \sqrt{x}\,\mathrm{d}\mu_\gamma(x)$. The common scale $\sigma\sqrt{n}$ cancels from RankMe, and the exact finite-d identity is

$$H_{\text{ent}}(\mathbf{p}) = \log d + \log S_d - \frac{1}{2S_d}\frac{1}{d}\sum_i \sqrt{\lambda_i}\log\lambda_i.$$

Consequently, with $0\log 0 = 0$, the Marchenko–Pastur limit gives

$$H_{\text{ent}}(\mathbf{p}) = \log d + \log m_{1/2} - \frac{1}{2m_{1/2}}\int \sqrt{x}\log x\,\mathrm{d}\mu_\gamma(x) + o(1).$$

The required moments are finite for every fixed positive aspect ratio, including regimes with support reaching zero or an atom at zero. Hence $r_{\text{eff}}(H)/d \to c(\gamma) > 0$ and $r_{\text{eff}}(H) = \Theta(d)$. □

Proof of Proposition 2. If $H = U\Sigma V^\top$, then $HQ = U\Sigma(Q^\top V)^\top$. Thus both retained bases transform by $V_T' = Q^\top V_T$ and $V_R' = Q^\top V_R$, so $(V_T')^\top V_R' = V_T^\top Q Q^\top V_R = V_T^\top V_R$. Singular values and the clipped score are also unchanged. To see that the target spectrum alone is insufficient, take a fixed rank-one reference along e_1 and a rank-one target with the same singular value but right direction $\cos\theta\, e_1 + \sin\theta\, e_2$. For $k_m = 1$, alignment is $\cos^2\theta$, which changes the score without changing the target spectrum. A tie crossing the truncation boundary requires a consistent subspace choice; arbitrary independent SVD tie-breaking is not covered by the invariance statement. □

Classwise total-variance diagnostic. On the VTAB architecture–task pairs for which the diagnostic is available, $\eta_\Sigma = \mathrm{Var}_c[\mathrm{tr}(\Sigma_c)]/(\mathbb{E}_c[\mathrm{tr}(\Sigma_c)])^2$ has median 0.022, mean 0.035, and 93.3% of values below 0.1. This measures similarity of total variance across classes, not directional isotropy: identical anisotropic class covariances can also give $\eta_\Sigma = 0$. We therefore do not use this statistic to validate the isotropic assumption of Lemma 1; that assumption defines an explanatory model rather than a verified property of every pretrained representation.

Width-plot protocol. The VTAB series uses uncentered RankMe singular-value ranks. Architecture means use eight cached datasets for this spectral diagnostic only; STL-10 is excluded from target selection. EX uses the same RankMe scores as its selection evaluation. MV is omitted from this architecture-mean plot because widths can differ within recorded grouping keys; its correlations are reported per variant.

B MV Pool (Multi-Variant Pool)

Dataset sources. CIFAR-10/100 originate from Krizhevsky [27], DTD from Cimpoi et al. [8], EuroSAT from Helber et al. [20], and SVHN from Netzer et al. [37]. ImageNet [11] is the pretraining dataset for the supervised candidates, not a target in the headline aggregate.

Setup. The MV inventory contains 132 pretrained variants in 48 recorded groups; the main architecture-selection support uses 119 variants in 35 groups with at least two members. Models are sourced from timm [54] and torchvision, spanning VGG, ResNet, DenseNet, EfficientNet, MobileNet, RegNet, ViT, DeiT, ConvNeXt, and Swin. The five tasks are CIFAR-10, CIFAR-100, DTD, EuroSAT, and SVHN. LP ground truth uses ℓ_2-regularized logistic regression ($C = 1$) on an $80/20$

split, as in VTAB. Table 11 reports actual recorded group counts rather than conflating broad backbone classes with grouping keys.

Architecture families and variant counts.

Table 11: Recorded MV architecture-group keys and variant counts on the fixed multi-variant support. There are 35 groups and 119 variants; broad classes such as ResNet contain several keys.

Group key	Variants	Group key	Variants
convnext_base	2	regnet	8
convnext_small	3	resnet101	4
convnext_tiny	3	resnet152	4
deit_base	2	resnet18	3
deit_small	2	resnet34	3
deit_tiny	2	resnet50	11
densenet121	3	resnext50	3
densenet169	3	swin_small	4
densenet201	3	swin_tiny	3
efficientnet_b0	3	tf_efficientnet	2
efficientnet_b2	3	vgg16	3
efficientnet_b3	3	vit_b32	2
efficientnet_b4	2	vit_base	9
inception_v3	3	vit_large	2
mixer_b16	2	vit_small	4
mnasnet	3	vit_tiny	2
mobilenet_v3_large	4	wide_resnet50_2	3
mobilenet_v3_small	3		

Width diagnostic. Per-task MV RankMe–width correlations range from 0.588 to 0.631 (mean 0.609). RankMe regret is 0.144, lower than random (0.1914) but higher than $\mathrm{SSR}^{\ddagger}$ (0.098). Removing VGG leaves both score-based regrets unchanged. On SVHN the oracle is DenseNet-169. EuroSAT has two tied oracle variants: DenseNet-201 and ViT-large, both with LP 0.955.

Multi-variant structure. The inventory has 132 variants in 48 groups; the headline support has 119 variants in 35 multi-variant groups. VGG16-BN is a singleton outside the headline support. Canonical representatives are the lexicographically first variant in each group. Flat evaluation uses 119 candidates. Full-132 diagnostic regrets are 0.152 for RankMe and 0.082 for $\mathrm{SSR}^{\ddagger}$, with their own oracle; they are separate from the headline comparison.

Provenance and LP sensitivity. Each torchvision VGG model has target and STL-10 caches for three extraction settings: 25088-dimensional flattened features, 4096 dimensions with two transforms, and 4096 dimensions with one official transform. Manifests record paths, timestamps, cache/weight/script hashes, LP splits, predictions, and solver versions. Independent checks verify all 36 caches and reproduce predictions for all 20 LP cells in the two 4096-dimensional settings. Those settings and the tested scikit-learn versions (1.6.1 and 1.2.2) give identical 22-unit RankMe/$\mathrm{SSR}^{\ddagger}$ regrets, though individual VGG LP accuracies can differ. This sensitivity test does not establish protocol equivalence for auxiliary metrics or untested candidates.

Version provenance and corrections. Earlier versions used flattened 25088-dimensional torchvision VGG features while describing 4096-dimensional penultimate features. The present inventory replaces the two VGG models' ten target rows; the other 650 rows are retained. The 22-unit SSR-minus-RankMe difference changes from -0.1043 to -0.0645, and MV RankMe is better than random. Earlier below-random MV claims, unsupported CNN-only/Transformer-only effects, and general within-family superiority statements are not supported. Effective-rank normalization, official label-dependent DiSCO, target-disjoint reference comparisons, and candidate supports are specified in the present protocols. Original affected cache provenance is incomplete; new manifests document replacement data rather than reconstructing that history.

C SSL/Multimodal Pool

Additional dataset sources. Besides the datasets cited in Appendices B, D, and E, the SSL inventory includes MNIST [28], USPS [22], GTSRB [48], Semeion [46], and Beans [34]. CIFAR-100-coarse uses the superclass labels of CIFAR-100 [27]. Rendered-SST2 is the image-rendered variant used by CLIP [44] of the Stanford Sentiment Treebank [47]. Beans and Semeion are credited through original dataset release records rather than an assumed standalone paper.

The eleven biomedical tasks use MedMNIST [57]: PathMNIST (colorectal histology [23]), DermaMNIST (HAM10000/ISIC [10, 52]), OCTMNIST and PneumoniaMNIST [24], RetinaMNIST (DeepDRiD [30]), BreastMNIST [3], BloodMNIST [1], TissueMNIST [32], and OrganAMNIST/OrganCMNIST/OrganSMNIST (LiTS images and organ annotations [5, 56]). The citations identify both the standardized tasks and their original image sources.

Setup. We extract penultimate-layer features from 8 SSL/multimodal encoders: ResNet-50/101, MoCo-v3-R50, ConvNeXt-B, ViT-B/16, CLIP-ViT-L/14, DINOv2-ViT-B/14, DINO-ViT-S/16. Feature dimensions $d \in \{384, 768, 1024, 2048\}$; no VGG-class architectures. Evaluation uses 29 downstream tasks covering VTAB, MedMNIST variants, and standard transfer benchmarks. LP ground truth uses the same $80/20$ logistic-regression protocol as MV pool.

Available support and scope. The retained SSL table has 209 of $8 \times 29 = 232$ possible model–task cells. There are eight candidates on 23 tasks, four on FGVC-Aircraft, five each on Flowers-102, Food-101, Oxford-Pets, and Rendered-SST2, and one on Semeion. The oracle is over available candidates; Semeion necessarily has zero selection regret. Official DiSCO scores match these same 209 LP cells. These historical diagnostics are not a complete rectangular benchmark and are excluded from the 22-unit headline comparison. The historical SSR script specifies ImageNet reference features with a per-model CIFAR-10 fallback, but a per-run reference manifest was not retained. We therefore do not treat its SSR columns as a verified target-disjoint STL-10 evaluation or use them to support the headline correction claim.

Diagnostic limitation. The incomplete SSL support and missing per-run reference manifest prevent a fully aligned RankMe/$\mathrm{SSR}^{\ddagger}$ comparison under the main target-disjoint protocol. Official DiSCO results remain descriptive on the available 209 cells and are separate from the headline zero-label comparison.

Label-dependent comparison. Official PCA-128 DiSCO gives regret 0.045 for $G = 6$ and 0.042 for $G = 8$ on the available 29 SSL tasks. Full-feature scoring gives $0.057/0.047$; these are different dimensionality settings. Official classification DiSCO uses target ground-truth labels for class-conditional covariances and is reported separately from the zero-label $\mathrm{SSR}^{\ddagger}$–RankMe comparison.

Table 7 in the main text reports these label-dependent results alongside MV, VTAB, and EX.

D VTAB Pool

Dataset sources. In addition to the shared sources in Appendix B, this pool uses Fashion-MNIST [55], Flowers-102 [39], Oxford-IIIT Pets [43], and STL-10 [9]. The names `pets` and `oxford_pets` denote the same dataset. "VTAB pool" names our retained evaluation pool, not a claim to reproduce the full original VTAB suite [59].

Setup. The final common pool contains the 24 timm-pretrained architectures with complete features and LP values on all 8 VTAB tasks (CIFAR-10/100, DTD, Fashion-MNIST, Flowers-102, Pets, STL-10, SVHN). The original manifest also listed ResNet-152, but only one of its eight feature files was available, so it is excluded uniformly rather than allowing the candidate set to vary by task. STL-10 features serve as the reference for SA_k and are excluded as a target in the common-support analysis.

Reference protocol and fixed k. The protocol uses uncentered SVD, RankMe's singular-value normalization, and excludes STL-10 as a target. With $k = 20$, the STL-10 single reference gives regret 0.061 over seven targets on the 24-architecture support. Reference sensitivity is evaluated only on target-disjoint comparisons. We use a fixed $k = 20$ without target-label tuning. Sensitivity to k has not been established under this final evaluation protocol.

Evaluation support. The inventories contain 660 MV, 510 EX, and 192 VTAB model–task cells; 168 VTAB cells correspond to non-reference targets. The 22-unit aggregate uses the multi-variant

MV support, all seven non-reference VTAB targets, and ten EX targets. The seven-task reference regrets are $0.061/0.055$ for the single reference and ensemble, respectively.

Multi-reference robustness. For a target-disjoint evaluation, we omit any reference whose dataset name equals the current target and take the median alignment across the remaining members of the predetermined set (STL-10, CIFAR-10, Flowers-102, Pets). This leave-one-target-out ensemble gives regret 0.055, versus 0.061 for STL-10 alone. Given the limited seven-task evidence and approximately M-fold reference-side cost, we retain a single reference as the default. Reference design remains an open problem.

E EX Pool and Stage-1 Comparisons

Dataset sources. FGVC-Aircraft [33] and Food-101 [6] supplement the eight target datasets credited in Appendices B and D; STL-10 is not an EX target.

EX pool (51-variant $\times$ 10-task, supervised). 51 ImageNet-1k pretrained variants from 35 canonical architectures (including VGG-16, d=4096; $d_{\max}/d_{\min}$=21.3×) on 10 target tasks spanning fine-grained recognition (FGVC-Aircraft, Flowers-102, Food-101, Oxford-Pets), texture/scene (DTD, EuroSAT), digit/fashion (SVHN, Fashion-MNIST), and general object (CIFAR-10/100); features extracted on the full dataset. Flat-$\mathrm{SSR}^{\ddagger}$ regret is 0.0595, compared with 0.1445 for RankMe. Official, target-label-dependent DiSCO obtains $+0.045$ with G=6 and $+0.049$ with G=8; it is not part of the zero-label comparison.

Table 5 gives the complete-support auxiliary comparison.

Spectral truncation. On the complete 118-candidate auxiliary support, top-256 RankMe has regret 0.134, between full RankMe (0.144) and $\mathrm{SSR}^{\ddagger}$ (0.098). The corresponding two-stage truncated-score→RankMe diagnostic has regret 0.136, versus 0.102 for $\mathrm{SSR}^{\ddagger}$ →RankMe. These are setting-specific observations; they do not establish that truncation is intrinsically unsuitable.

F Ablations and Negative Results

Information budgets. DiSCO requires target labels; $\mathrm{SSR}^{\ddagger}$ does not. Width range is not a validated selection rule. The default relational weight is one, without cross-pool tuning.

Algebraic form of the correction. On MV, $-\alpha \log \mathrm{SA}_k$ gives regrets 0.098, 0.098, and 0.093 at the tested $\alpha = 0.5, 1, 2$, respectively. These three points do not establish stability over a continuous interval. On the 118-candidate support, the additive variant $g(s) = 1 - s$, $\beta = 1$ gives regret 0.117. The implementation $(\log r_{\mathrm{eff}} - s)/\log C$ omits the task-constant $1/\log C$; its ranking and regret, but not raw scores, equal the family definition. The inverted-direction control $-\log(1 - \mathrm{SA}_k)$ gives 0.388. We keep $\alpha = 1$ without target-based tuning; these controls show sensitivity to the form of the relational term.

Per-task selection. Table 4 gives the selections and oracles. RankMe selects a ResNeXt variant on four tasks and Inception on SVHN. $\mathrm{SSR}^{\ddagger}$ wins four paired task comparisons but loses on SVHN.

UID-normalized score. $R_{\mathrm{UID}} = \log r_{\mathrm{eff}}/(\mathrm{UID} \log C)$ divides log effective rank by the TwoNN intrinsic-dimension estimate UID [14] and $\log C$, rather than ranking candidates directly by UID. On 118 complete candidates its flat regret is 0.364 (versus $\mathrm{SSR}^{\ddagger}$'s 0.098), but R_{UID} →RankMe gives 0.058 on both this support and the headline 119: all 35 canonical scores exist despite one missing noncanonical auxiliary cell. Poor flat ranking need not imply poor screening. This post-hoc five-task result outperforms the evaluated $\mathrm{SSR}^{\ddagger}$ pipelines, without establishing a general UID-based selection rule.

Narrow-pool scope. The available evidence does not identify a universally preferable score from CNN/Transformer membership alone. Claims about homogeneous sub-pools require matched candidate and task support.

Within-family ranking and task scaling. Excluding two constant-LP groups from 35×5 architecture–task groups leaves 173 with mean Spearman correlations 0.527 (RankMe) and 0.519

($\mathrm{SSR}^{\ddagger}$): descriptive, not significant superiority evidence. Common positive task scaling, including $1/\log C(T)$, preserves candidate ordering. We aggregate per-task regrets, not raw scores.

G Experimental Environment and Compute Accounting

NPU feature extraction. The MV/EX DiSCO feature-extraction launcher uses PyTorch with `torch_npu`/Ascend, four NPU devices (0–3), one model shard per device, batch size 64, inference mode, and float32 caches. Hardware provenance is not complete for every cached feature file.

Cached-feature analysis. DiSCO, linear probes, reference comparisons, and bootstrap use CPU-based NumPy/SciPy/scikit-learn routines. The MV/EX DiSCO launcher uses 16 workers; caching avoids repeating encoder inference for each score.

VGG extraction environment. The two VGG models use an NVIDIA L20 (about 46 GB), batch size 64, float32 caches, PyTorch 2.5.1+cu124, torchvision 0.20.1, Python 3.11.11, NumPy 1.26.4, SciPy 1.15.3, and scikit-learn 1.6.1. The host exposes 16 CPU threads and about 123 GiB RAM; scoring is capped at eight threads. LP uses seed-0 stratified $800/200$ splits, train-only StandardScaler, $C = 1$, lbfgs, and max_iter=1000. An isolated sklearn 1.2.2 comparison changes some VGG LPs but not the headline regrets. Cache and script hashes and generation timestamps are recorded.

Reporting scope. Section 6 counts candidate–task evaluations, not runtime, energy, or hardware speedups; reference setup is additional. Incomplete historical hardware, software, and runtime records preclude full compute reporting (Checklist: "No").